

AMENDMENT NO. 1 JUNE 1999
TO
IS 2386 (PART VII) 1963 METHODS OF TEST
FOR AGGREGATES FOR CONCRETE
PART VII ALKALI AGGREGATE REACTIVITY

(*Page 15, clause 2.10*) — Insert the following new clauses after 2.10:

2.11 For aggregates, though not containing known deleterious silica minerals like opal, chalcedony, tridymite and cristobolite, xerolite, heulandite, glassy to crypto-crystalline rhyolites, decites and andesites and their tuffs and certain phyllites but containing more than 20 percent strained quartz undulatory extinction angle greater than 15° causing deleterious reaction [see IS 2386 (Part VIII) : 1963] and also possibly showing presence of microcrystalline quartz, additional mortar-bar test with conditions as given in 2.11.1 and 2.11.2 keeping other condition same, shall be conducted.

2.11.1 The specimens shall be placed in the metal container only and the storage temperature in the container shall be $60 \pm 2^{\circ}\text{C}$.

2.11.2 Ordinary Portland cement whose alkali content is 1.0 percent or more (expressed as $\text{Na}_2\text{O} + 0.658 \text{ K}_2\text{O}$) shall be used for evaluation instead of 0.6 percent specified in Note 1 under 2.4.3, other cement aggregate combination specified under 2.4.3 remaining the same.

2.12 It is recommended that strained quartz bearing aggregates showing 180 day expansions of 0.06 percent using ordinary Portland cement with 1.0 percent or more alkali content as per 2.11.2 be classified as potentially reactive.

NOTES

1 for coarse aggregate of composite nature, the test shall be conducted on the composite sample.

2 The test is of accelerated nature and suitable for strained quartz bearing aggregates only which are slow-reactive in nature. The test is not suitable for aggregates containing conventional known deleterious silica minerals mentioned above.'

(CED 2)